Q1. JEE Main 2026 (21 January Shift 2)

Given below are two statements :

Statement I : The correct order in terms of atomic/ionic radii is $Al > Mg > Mg^{2+} > Al^{3+}$.

Statement II : The correct order in terms of the magnitude of electron gain enthalpy is $Cl > Br > S > O$.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are false

(2) Both Statement I and Statement II are true

(3) Statement I is false but Statement II is true

(4) Statement I is true but Statement II is false

Q2. JEE Main 2026 (23 January Shift 1)

The correct trend in the first ionization enthalpies of the elements in the 3rd period of periodic table is :

(1) $Si < S < Al < P < Cl$

(2) $Al < S < P < Si < Cl$

(3) $Al < Si < S < P < Cl$

(4) $S < Si < Al < P < Cl$

Q3. JEE Main 2026 (28 January Shift 2)

Consider the elements N, P, O, S, Cl and F. The number of valence electrons present in the elements with most and least metallic character from the above list is respectively.

(1) 7 and 5

(2) 5 and 7

(3) 5 and 6

(4) 6 and 7

Q4. JEE Main 2026 (24 January Shift 2)

The correct order of C, N, O and F in terms of second ionisation potential is

(1) $C < N < F < O$

(2) $C < O < N < F$

(3) $C < F < N < O$

(4) $F < N < C < O$

Q5. JEE Main 2026 (24 January Shift 2)

Choose the INCORRECT statement

(1) CO_2 is the most acidic oxide among the dioxides of group of 14 elements.

(2) Carbon exhibits negative oxidation states along with +4 and +2.

(3) Carbon cannot exceed its covalency more than four.

(4) Among the isotopes of carbon, ^{13}C is a radioactive isotope.

Q6. JEE Main 2026 (22 January Shift 2)

Given below are two statements :

Statement I: $C < O < N < F$ is the correct order in terms of first ionization enthalpy values.

Statement II: $S > Se > Te > Po > O$ is the correct order in terms of the magnitude of electron gain enthalpy values.

In the light of the above statements, choose the correct answer from the options given below :

(1) Statement I is true but Statement II is false
(2) Statement I is false but Statement II is true
(3) Both Statement I and Statement II are false
(4) Both Statement I and Statement II are true

Q7. JEE Main 2026 (22 January Shift 1)

Given below are two statements:

Statement I: The halogen that makes longest bond with hydrogen in HX, has the smallest covalent radius in its group.

Statement II: A group 15 element's hydride EH_3 has the lowest boiling point among corresponding hydrides of other group 15 elements. The maximum covalency of that element E is 4.

In the light of the above statements, choose the correct answer from the options given below

(1) Both Statement I and Statement II are true
(2) Both Statement I and Statement II are false
(3) Statement I is false but Statement II is true
(4) Statement I is true but Statement II is false

Q8. JEE Main 2026 (23 January Shift 2)

Elements X and Y belong to Group 15. The difference between the electronegativity values of ' X ' and phosphorus is higher than that of the difference between phosphorus and ' Y '. ' X ' \& ' Y ' are respectively

(1) N & As
(2) As & Sb
(3) Bi & N
(4) As & Bi

Q9. JEE Main 2026 (23 January Shift 2)

Given below are two statements:

Statement I: The second ionisation enthalpy of Na is larger than the corresponding ionisation enthalpy of Mg.

Statement II: The ionic radius of O^{2-} is larger than that of F^{-}.

In the light of the above statements, choose the correct answer from the options given below

(1) Statement I is false but Statement II is true
(2) Both Statement I and Statement II are false
(3) Statement I is true but Statement II is false
(4) Both Statement I and Statement II are true

Q10. JEE Main 2026 (21 January Shift 1)

Which of the following represents the correct trend for the mentioned property ?

A. $F > P > S > B$ - First Ionization Energy

B. $Cl > F > S > P$ - Electron Affinity

C. $K > Al > Mg > B$ - Metallic character

D. $K_2O > Na_2O > MgO > Al_2O_3$ - Basic character

Choose the correct answer from the options given below :

(1) B and C only
(2) A, B and D only
(3) A and B only
(4) A, B, C and D

Q11. JEE Main 2026 (28 January Shift 1)

In period 4 of the periodic table, the elements with highest and lowest atomic radii are respectively.

(1) Rb & Br
(2) K & Se
(3) K & Br
(4) Na & Cl

Q12. JEE Main 2026 (24 January Shift 1)

Given below are two statements:

Statement I: $K > Mg > Al > B$ is the correct order in terms of metallic character.

Statement II: Atomic radius is always greater than the ionic radius for any element.

In the light of the above statements, choose the correct answer from the options given below

(1) Both Statement I and Statement II are true
(2) Both Statement I and Statement II are false
(3) Statement I is false but Statement II is true
(4) Statement I is true but Statement II is false

ANSWERS AND SOLUTIONS

1. (3)	**2.** (3)	**3.** (2)	**4.** (1)	**5.** (4)	**6.** (4)	**7.** (2)	**8.** (1)
9. (4)	**10.** (2)	**11.** (3)	**12.** (4)				

1. **(3)** Analyzing Statement I: The order given is Al > Mg > Mg^{2+} > Al^{3+}. However, the correct atomic radii order should be Mg > Al because Mg is in group 2 (period 3) while Al is in group 13 (period 3), and moving right across a period decreases radius. The correct order would be Mg > Al > Mg^{2+} > Al^{3+}. Therefore, Statement I is FALSE.

Analyzing Statement II: Electron gain enthalpy magnitudes follow: Cl (highest, ~349 kJ/mol) > Br (~325) > S (~200) > O (~141). This is because Cl has the highest tendency to gain an electron due to its position and electronic configuration. Statement II is TRUE.

Since Statement I is false and Statement II is true, the answer is option (3).

2. **(3)** First ionization enthalpy generally increases across a period due to increasing nuclear charge and decreasing atomic radius. However, exceptions occur based on electron configuration stability.
For 3rd period elements, the approximate ionization enthalpy values are: Al (578), Si (787), P (1012), S (1000), Cl (1251) kJ/mol.
(1) Al has lower IE than Si because removing $3p^1$ is easier than removing one electron from $3p^2$.
(2) S has slightly lower IE than P despite being to the right because P has a half-filled p-orbital ($3p^3$) which is more stable.
(3) Cl has the highest IE as it approaches a filled p-orbital.
The correct order is: Al < Si < S < P < Cl

3. **(2)** Valence electrons: N(5), P(5), O(6), S(6), Cl(7), F(7).
Metallic character order in the list: P > S > N > Cl > O > F.
Most metallic character: P with 5 valence electrons.
Least metallic character (most nonmetallic): F with 7 valence electrons.
F is the most electronegative and has the smallest atomic size, making it most nonmetallic.

4. **(1)** Second ionization potential (I_2) is the energy to remove the second electron.
After removing the first electron: C^+ has $2p^1$, N^+ has $2p^2$, O^+ has $2p^3$ (half-filled, stable), F^+ has $2p^4$.
C has lowest I_2 as 2p electrons are least tightly bound with lowest nuclear charge.
N has slightly higher I_2 than C due to increased nuclear charge.
F has highest I_2 due to highest effective nuclear charge.
O^+ with half-filled 2p is relatively stable, making I_2 of O intermediate.
Order: $C < N < F < O$

5. **(4)** Analyzing each statement:

1 CO_2 is most acidic among group 14 dioxides - correct, it's a non-metal acidic oxide.

2 Carbon shows negative oxidation states (-4 in methane, -3 in ethane, etc.) along with +2 and +4 - correct.

3 Carbon covalency cannot exceed 4 - correct, as it's period 2 element unable to expand octet.

4 ^{13}C is radioactive - incorrect. ^{13}C is a stable, non-radioactive isotope. ^{14}C is the radioactive isotope used in radiocarbon dating with half-life of 5730 years. Therefore, statement (4) is incorrect.

6. **(4)** Statement I:

Ionization enthalpy generally increases across a period from left to right.

However, Nitrogen ($2s^2 2p^3$) has a stable half-filled electronic configuration, making it harder to remove an electron compared to Oxygen ($2s^2 2p^4$).

Thus, the order is $C < O < N < F$.

Statement I is true.

Statement II:

For Group 16 elements, Oxygen has an exceptionally low magnitude of electron gain enthalpy due to its small size and high inter-electronic repulsions in the $2p$ subshell.

The magnitude of electron gain enthalpy decreases down the group from Sulphur onwards.

The correct order of magnitude is $S > Se > Te > Po > O$.

Statement II is true.

Both statements are correct.

7. **(2)** $HF < HCl < HBr < HI$ (bond length order)

$F < Cl < Br < I$ (radius order)

$PH_3 < AsH_3 < NH_3 < SbH_3 < BiH_3$ (Boiling point order)

Maximum possible covalency of phosphorous is 6

8. **(1)** Group 15 elements have electronegativity values (Pauling scale): N = 3.04, P = 2.19, As = 2.18, Sb = 2.05, Bi = 2.02.

The condition requires: |EN(X) - EN(P)| > |EN(P) - EN(Y)|.

(1) N and As: |3.04 - 2.19| = 0.85 and |2.19 - 2.18| = 0.01. Since 0.85 > 0.01, the condition is satisfied.

(2) As and Sb: |2.18 - 2.19| = 0.01 and |2.19 - 2.05| = 0.14. Since 0.01 < 0.14, it fails.

(3) Bi and N: |2.02 - 2.19| = 0.17 and |2.19 - 3.04| = 0.85. Since 0.17 < 0.85, it fails.

(4) As and Bi: |2.18 - 2.19| = 0.01 and |2.19 - 2.02| = 0.17. Since 0.01 < 0.17, it fails.

9. **(4)** Statement I examines second ionization enthalpy:

Na ($1s^2\ 2s^2\ 2p^6\ 3s^1$) loses its first electron to achieve a stable noble gas configuration. The second ionization would break into the 2p core, requiring enormous energy.

Mg ($1s^2\ 2s^2\ 2p^6\ 3s^2$) loses its second electron from the 3s orbital, which is less stable.

Thus, IE_2 of Na is much larger than IE_2 of Mg.

Statement I is TRUE.

Statement II compares O^{2-} and F^-:

both are isoelectronic with 10 electrons.

O^{2-} has 8 protons while F^- has 9 protons.

The lower nuclear charge in O^{2-} results in weaker electrostatic attraction, making the ionic radius of O^{2-} larger than F^-.

Statement II is TRUE.

10. **(2)** A. F > P > S > B (IE_1): F has highest IE_1, P > S due to half-filled $3p^3$ stability, B has lowest. Correct

B. Cl > F > S > P (EA): Cl has highest EA (F smaller, more repulsion), S > P (half-filled p-orbital). Correct

C. K > Al > Mg > B (Metallic character): Incorrect. Mg (Group 2) is more metallic than Al (Group 13) in same period. Correct order: K > Mg > Al > B

D. $K_2O > Na_2O > MgO > Al_2O_3$ (Basic character): Follows electropositive nature. Al_2O_3 is amphoteric. Correct

Correct statements: A, B and D only.

11. **(3)** In period 4, atomic radii decrease from left to right as nuclear charge increases while electron shell remains constant. K (potassium) is at the extreme left of period 4 metals and has the highest atomic radius. Br (bromine) is near the right end as a nonmetal and has the lowest atomic radius in this period. Therefore, the elements with highest and lowest atomic radii in period 4 are K and Br respectively.

12. **(4)** Statement I: Metallic character decreases from left to right across a period and increases down a group. K (Group 1, Period 4) is the most metallic element among these. Mg (Group 2, Period 3) is less metallic than K. Al (Group 13, Period 3) is less metallic than Mg. B (Group 13, Period 2) is a metalloid with the least metallic character. The order K > Mg > Al > B correctly represents decreasing metallic character. TRUE.

Statement II: Atomic radius compared to ionic radius depends on whether the ion is a cation or anion. For metals forming cations: atomic radius > cationic radius (e.g., Na > Na^+). However, for nonmetals forming anions: atomic radius < anionic radius (e.g., Cl < Cl^-). The statement claims atomic radius is ALWAYS greater for ANY element, which is incorrect because anions have larger radii than their parent atoms. FALSE.

Correct answer: Statement I is true but Statement II is false.